🌀 M003 Extension — The Loop Spiral (Advanced)

Topic: One loop that **turns** (x, y, z) · **Course:** 3D Coordinates · **Level:** Extension (Advanced, after the Loop Staircase) · **Time:** about 45 minutes

The **staircase** took the **same** step every pass, so it grew in a **straight diagonal**. A **spiral** also **turns** every pass — same loop, but the direction **changes** — so the squares wrap into a tower. `step % 4` picks the direction; `y = y + 1` climbs.

```
x = 8
y = 0
z = 8
for step in range(12):
    fill GOLD from (x, y, z) to (x + 2, y, z + 2)
    y = y + 1
    if step % 4 == 0:
        x = x + 4
    elif step % 4 == 1:
        z = z + 4
    elif step % 4 == 2:
        x = x - 4
    else:
        z = z - 4
```

> 🔁 **Big idea:** the `fill` line **never changes**. The tower spirals only because **x, z change direction** each pass and **y rises**. Same code, new numbers.

	3 ↑ back **z − 4**	
2 ← left **x − 4**	**step % 4**	**0 → right** **x + 4**
	1 ↓ forward **z + 4**	

Trace (or **dry-run**) means: be the computer. Walk the code line by line and write what each square uses **before** you run it.

1 · Predict 🔮

Trace the loop. For each step write the **turn**, and the **x, z** the `fill` line uses. `step % 4` is filled for you. Step 0 is done. The turn changes x or z for the **next** row.

step	step % 4	turn	x used	z used
0	0	right	8	8
1	1			
2	2			
3	3			
4	0			
5	1			
6	2			
7	3			

The 12 squares use only 4 **different (x, z) spots. Why? Explain using the pattern** `0, 1, 2, 3`**.**

Which line makes each square *climb*? Which two-part rule makes it *turn*? Say why for each.

2 · Spot the Bug 🐛

Each loop is broken. Trace it, fix it, and explain **why**.

Bug A — no turn. The four `if` / `elif` lines were deleted, so the direction never changes.

```
for step in range(12):
    fill GOLD from (x, y, z) to (x + 2, y, z + 2)
    y = y + 1
    x = x + 4
```

What shape do you get instead of a spiral, and why? Write the fix.

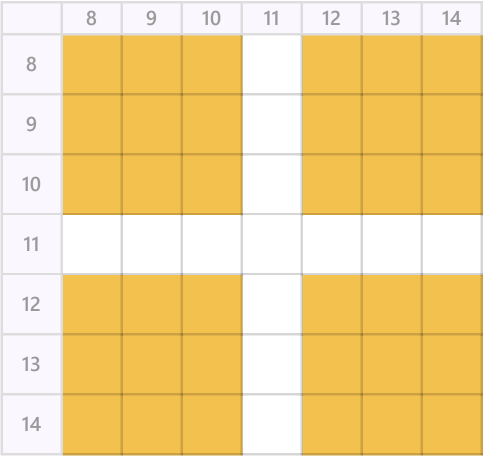
Bug B — turn too small. A friend changed each turn to `+ 2` and `- 2`. It runs, but the four corners overlap into a blob.

Each square is 3 wide. Explain why a turn of `+ 4` clears the last square but `+ 2` overlaps it.

3 · Show Your Work 📸 🎥

Load the world. Stand on your home spot. Type `SPIRAL` and watch each square **turn** to a new corner and **climb**.

👁 Top-down view (looking DOWN): x → across, z ↓ deeper



Looking down, the squares sit on 4 corners — the loop turns right, forward, left, back, then repeats, climbing each time.

Modify challenge. Change **one** thing and **predict before you run** — e.g. `range(12)` → `range(20)` (taller), or each turn `+ 4` → `+ 6` (wider). Write what will change, then build it.

Level up — the ROUND spiral. The square spiral turns at **4 sharp corners**. To make a **smooth round** spiral, move around a **circle** with `cos` and `sin` instead:

```
radius = 6
for step in range(40):
    angle = step * 0.5
    x = round(radius * cos(angle))
    z = round(radius * sin(angle))
    fill GOLD from (x, step, z) to (x + 2, step, z + 2)
```

🧭 `cos` and `sin` trace a **circle**; `step` as the height makes the circle **climb** — a smooth round spiral (a helix). `angle` grows every pass, so the direction keeps turning.

Predict: will the round spiral have corners or a smooth curve? What does a bigger `radius` change?

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what the loop does and how the turn works.

2 · Your build. Point the camera at the spiral. Show it climbing and turning.

Fill the blanks:

> Today I built ______.
>
> I built it using this code: ______.
>
> In this code I used ______.
>
> The hardest part was ______.
>
> That part was hard because ______.
>
> The most fun part was ______.
>
> Something new I learned was ______.

Write your lines here, then say them in your video.

Submit ✅

Send this worksheet + your **one** video to teacher on KakaoTalk.